

SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE
SUBJECT - Data Analysis with R/SAS/SPSS

Aim - Creating cross-tabulations and two-way tables using table() (R).

Output :

The first screenshot shows the RStudio interface with the following code in the script editor:

```
27 table(data$volatile.acidity, data$citric.acid)
28
29 # Volatile Acidity vs Residual Sugar
30 table(data$volatile.acidity, data$residual.sugar)
31
32 # Citric Acid vs Residual Sugar
33 table(data$citric.acid, data$residual.sugar)
34
35 # -----
36 # Optional: Add margins (row & column totals)
37 #
38
39 addmargins(table(data$fixed.acidity, data$volatile.acidity))
40
41
```

The console output shows the results of the first two table() functions and the head of the data frame:

```
> # Clear workspace
> rm(list = ls())
> # Load the dataset
> data <- read.csv("winequality.csv")
> # View first few rows
> head(data)
  fixed.acidity volatile.acidity citric.acid residual.sugar chlorides free.sulfur.dioxide total.sulfur.dioxide density pH
1          7.0             0.27         0.36          20.7         45.00             45             170       1.001 3.00
2          6.3             0.30         0.34           1.6         49.00             14             132     994.000 3.30
3          8.1             0.28         0.40           6.9          0.05             30              97      9.951 3.26
4          7.2             0.23         0.32           8.5         58.00             47             186     9.956 3.19
5          7.2             0.23         0.32           8.5         58.00             47             186     9.956 3.19
6          8.1             0.28         0.40           6.9          0.05             30              97      9.951 3.26
  sulphates quality alcohol
1    0.45      6 R$ 45.512,00
2    0.49      6 R$ 45.421,00
3    0.44      6 R$ 45.301,00
4    0.40      6 R$ 45.544,00
5    0.40      6 R$ 45.544,00
6    0.44      6 R$ 45.301,00
```

The second screenshot shows the same RStudio interface with the following code:

```
27 table(data$volatile.acidity, data$citric.acid)
28
29 # Volatile Acidity vs Residual Sugar
30 table(data$volatile.acidity, data$residual.sugar)
31
32 # Citric Acid vs Residual Sugar
33 table(data$citric.acid, data$residual.sugar)
34
35 # -----
36 # Optional: Add margins (row & column totals)
37 #
38
39 addmargins(table(data$fixed.acidity, data$volatile.acidity))
40
41
```

The console output shows the results of the first two table() functions and the str() function output:

```
> # Check structure
> str(data)
'data.frame': 4898 obs. of 12 variables:
 $ fixed.acidity : num 7 6.3 8.1 ...
 $ volatile.acidity : num 0.27 0.3 0.28 ...
 $ citric.acid : num 0.36 0.32 0.4 ...
 $ residual.sugar : num 20.7 1.6 49.0 ...
 $ chlorides : num 45 14 30 ...
 $ free.sulfur.dioxide : num 45 14 30 ...
 $ total.sulfur.dioxide : num 170 132 97 ...
 $ density : num 1.001 994.000 9.951 ...
 $ pH : num 3.00 3.30 3.26 ...
 $ sulphates : num 0.45 0.49 0.44 ...
 $ quality : int 6 6 6 ...
 $ alcohol : chr "R$ 45.512,00" "R$ 45.421,00" "R$ 45.301,00" "R$ 45.544,00" ...
```

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